

SUBTRACTION PUZZLES

Let's Solve the Mystery!

Subject:	Mathematics	Grade:	2nd Grade
Duration:	60 Minutes	Theme:	Mystery / Puzzle Detective

Welcome to your mission, Maths Detectives! This guide will help you unpack the secrets of subtraction using real-world clues like delicious gulab jamuns, rupee coins, and juicy mangoes. Let's solve the puzzles together!

1. Learning Outcomes

GOAL	FOCUS AREA	WHAT YOU WILL LEARN
LO 1	Conceptual Understanding	Explain what subtraction means (taking away / finding how many are left) and identify key vocabulary.
LO 2	Vocabulary Mastery	Use the correct terms: minuend, subtrahend, difference, minus, remaining.
LO 3	Puzzle Solving - Type A	Calculate the missing difference when minuend and subtrahend are known.
LO 4	Puzzle Solving - Type B	Find the missing subtrahend using reverse thinking (subtraction or addition).
LO 5	Puzzle Solving - Type C	Determine the missing minuend by applying the addition-helps strategy.
LO 6	Strategy Application	Use number-line hops and the inverse-operation trick to verify answers.

2. Key Vocabulary

TERM	MEANING
Subtraction	The operation of taking away one number from another to find how many remain.
Minuend	The BIG starting number in a subtraction sentence (the number we subtract FROM).
Subtrahend	The number being taken away or subtracted.
Difference	The result or answer of a subtraction problem; what is left.
Minus (-)	The symbol that represents the subtraction operation.
Take Away	A phrase meaning subtract; describes removing objects from a group.
Remaining	What is left after subtraction has been performed.
Number Line	A visual tool; hop BACKWARD along it to model subtraction.
Inverse Operation	The opposite operation; addition is the inverse of subtraction.



Story Hook: The Mystery of the Missing Gulab Jamuns

Ramu made 20 delicious gulab jamuns for his family. But when he came back, some had disappeared mysteriously! Put on your detective hats and use your subtraction skills to crack this mystery!

3. Parts of a Subtraction Sentence

Every subtraction sentence is made up of three specific parts. Take a look at this example:

$$15 - 6 = 9$$

Priya had 15 laddoos and gave 6 to Raju. 9 laddoos remain.

TERM	POSITION IN EQUATION	EXAMPLE ($18 - 5 = 13$)
Minuend	The first (large) number	18
Subtrahend	The number subtracted	5
Difference	The answer / result	13

4. The Three Puzzle Types

Type A: Missing Difference Easiest

Both the minuend and the subtrahend are known. Simply perform the subtraction to find what is left.

$$18 - 5 = _ \rightarrow \text{Answer: } 13$$

Story Case: Meena had 18 rotis. She served 5 to her family. How many rotis are left?

Type B: Missing Subtrahend Medium

The minuend and the difference are known; you must figure out how much was taken away. Think backwards: if $20 - ? = 13$, ask yourself "what do I add to 13 to get 20?"

$$20 - _ = 13 \rightarrow \text{Answer: } 7 \text{ (because } 13 + 7 = 20\text{)}$$

Story Case: Rahul had 20 rupee coins. He spent some at the shop. Now he has 13 coins left. How many rupees did he spend?

Type C: Missing Minuend Most Challenging

The subtrahend and the difference are known; your task is to find the original starting number. Connect them by adding the subtrahend to the difference.

$$_ - 6 = 10 \rightarrow \text{Answer: } 16 \text{ (because } 10 + 6 = 16\text{)}$$

Story Case: Sunita had some mangoes. She gave 6 away. Now she has 10. How many did she have at first?

At-a-Glance Summary Table

PUZZLE TYPE	EQUATION EXAMPLE	KEY STRATEGY	ANSWER
Type A (Missing Difference)	$18 - 5 = \underline{\hspace{1cm}}$	Subtract directly; count back on number line	13
Type B (Missing Subtrahend)	$20 - \underline{\hspace{1cm}} = 13$	Think backwards; use addition inverse	7
Type C (Missing Minuend)	$\underline{\hspace{1cm}} - 6 = 10$	Add difference + subtrahend to find start	16

5. Key Core Strategies

A. Reverse Thinking: Addition Helps!

When a piece is missing from a subtraction sentence, you can flip it to an addition equation to find the answer. This works perfectly because addition and subtraction are inverse operations.

SUBTRACTION PUZZLE	ADDITION EQUATION CONNECTION	FINAL ANSWER
$\underline{\hspace{1cm}} - 6 = 10$	$10 + 6 = \underline{\hspace{1cm}}$	16
$20 - \underline{\hspace{1cm}} = 13$	$13 + \underline{\hspace{1cm}} = 20$	7

Real-life story: Amma bought bananas. She gave 8 to guests. 12 bananas are left. How many did she buy?

→ $12 + 8 = 20$ bananas.

B. Number Line Strategy: Hop Backward!

The number line gives you a clear visual look at subtraction. Start at the **minuend** and hop backward by the **subtrahend** value. Wherever you land is your **difference**.

Example: Start at 15, hop back 4 steps, and land on 11. → $15 - 4 = 11$

6. Lesson Challenges & Activities

- **Warm-Up Mystery:** If Ramu started with 20 gulab jamuns and 7 disappeared, how many remain? ($20 - 7 = 13$).

- **Partner Game:** Take turns posing Type-B puzzles to each other. One partner writes a subtraction puzzle, and the other flips it to addition to solve it.
- **Crack the Code Challenge:** Solve 5 subtraction puzzles (a mix of Types A, B, and C). Each answer unlocks a letter. Once put together, the letters spell a secret Indian word (like **MANGO** or **DIWALI**)!

7. Guided Practice & Problems

PROBLEM	PUZZLE TYPE	WORKING & THINKING	ANSWER
$25 - 8 = \underline{\quad}$	Type A	Subtract directly: $25 - 8$	17
$35 - \underline{\quad} = 18$	Type B	Think: $18 + \underline{\quad} = 35 \rightarrow 35 - 18$	17
$\underline{\quad} - 9 = 15$	Type C	Think: $15 + 9 = \underline{\quad}$	24
$42 - 15 = \underline{\quad}$	Type A (Quiz)	Subtract directly: $42 - 15$	27
$40 - \underline{\quad} = 25$	Type B (Quiz)	Think: $25 + \underline{\quad} = 40 \rightarrow 40 - 25$	15
$\underline{\quad} - 12 = 20$	Type C (Quiz)	Think: $20 + 12 = \underline{\quad}$	32
$30 - \underline{\quad} = 14$	Type B (Story)	Chotu's pens: $14 + \underline{\quad} = 30 \rightarrow 30 - 14$	16

8. Extension & Homework

Home Mystery Task:

Ask a family member to hide some small objects (like spoons or chapatis). Write your own subtraction mystery in your maths journal using this framework:

"There were _____ objects at first. _____ disappeared. Now there are _____ left."

Bring your custom mystery to class tomorrow!

 You Cracked the Mystery! You're Official Maths Detectives! 